



ecology

Name:

Class:

Date:

Time:

Marks:

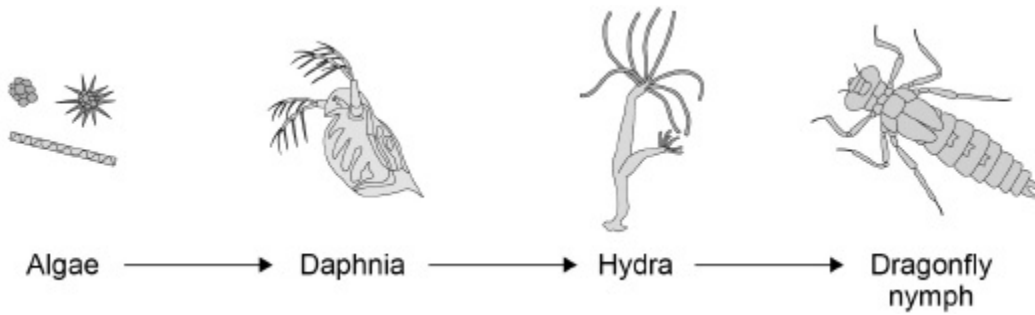
62 marks

Comments:

1.

Figure 1 shows a food chain in a pond.

Figure 1



(a) Which term describes the Daphnia in this food chain?

Tick (✓) **one** box.

Apex predator

☐

Primary consumer

☐

Producer

☐

Secondary consumer

☐

(1)

(b) Draw a pyramid of biomass for the food chain.

Label each trophic level.

(2)

(c) Give **one** reason why the total biomass of the Daphnia in the pond is different from the total biomass of the algae.

(1)

Students investigated the size of the population of Daphnia in the pond.

This is the method used.

1. Collect 1 dm³ of pond water from near the edge of the pond.
2. Pour the water through a fine net.
3. Count the number of Daphnia caught in the net.
4. Repeat steps 1–3 four more times.

The table below shows the results.

Sample number	Number of Daphnia in 1 dm ³ water
1	5
2	21
3	0
4	16
5	28

- (d) Calculate the mean number of Daphnia in 1 m³ of pond water.

$$1 \text{ m}^3 = 1000 \text{ dm}^3$$

Mean number of Daphnia in 1 m³ of pond water = _____

(2)

(e) The pond was a rectangular shape, measuring:

- length = 2.5 metres
- width = 1.5 metres
- depth = 0.5 metres.

Calculate the estimated number of Daphnia in the pond.

Use your answer from part (d).

Give your answer in standard form.

Number of Daphnia in the pond = _____

(4)

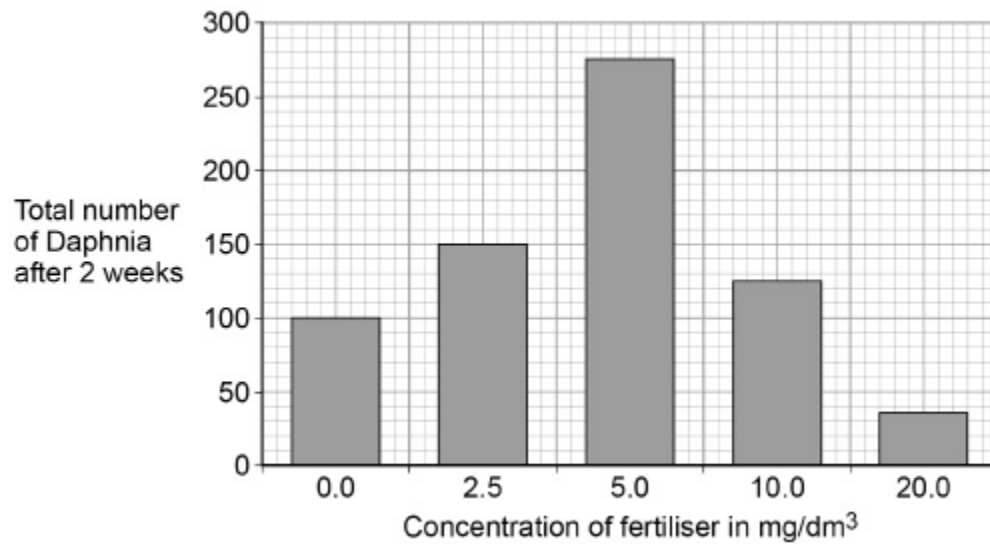
Rainfall can cause fertiliser to be washed from farmland into a pond.

The students investigated the effect of fertiliser on the population of Daphnia in water from the pond.

- The students put 20 Daphnia in each of five different concentrations of fertiliser.
- The students counted the total number of Daphnia in each concentration of fertiliser after 2 weeks.

Figure 2 shows the results.

Figure 2



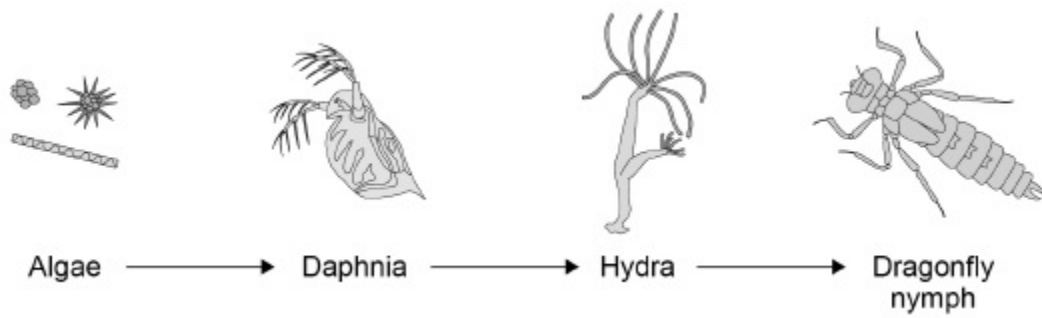
- (f) A concentration of 5.0 mg/dm³ of fertiliser caused a large increase in the population of Daphnia.

Explain why.

(2)

(g) **Figure 1** is repeated below.

Figure 1



The population of **Hydra** will decrease when 20 mg/dm^3 of fertiliser is added to the pond.

Explain why.

(2)

(Total 14 marks)

2.

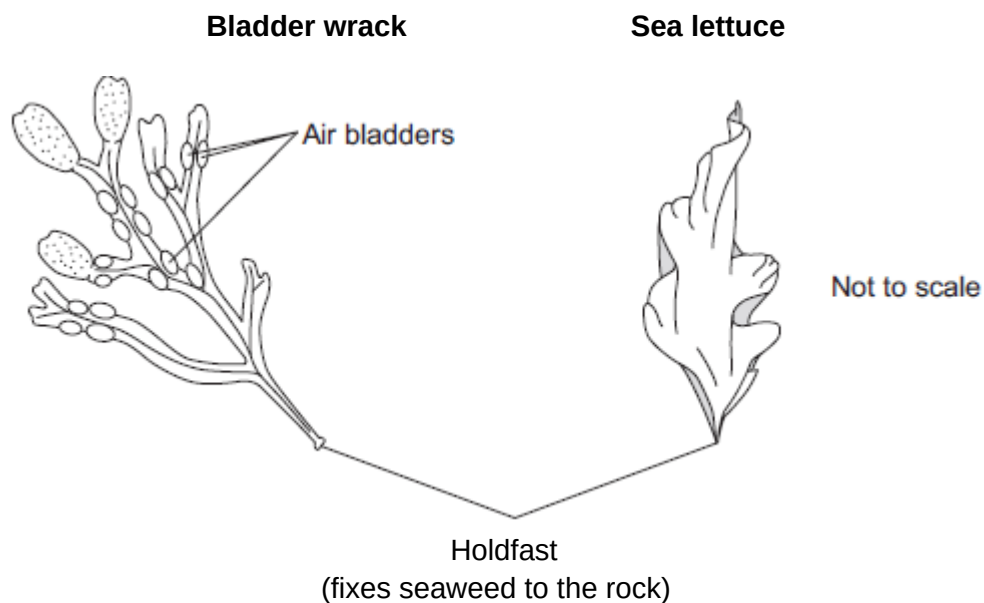
At the seashore, the tide comes in and goes out twice each day.

Some students investigated whether two different species of seaweed could live only at certain positions on a rocky shore.

Seaweeds are plant-like organisms that make their food by photosynthesis.

Figure 1 shows the two species of seaweed that the students investigated.

Figure 1

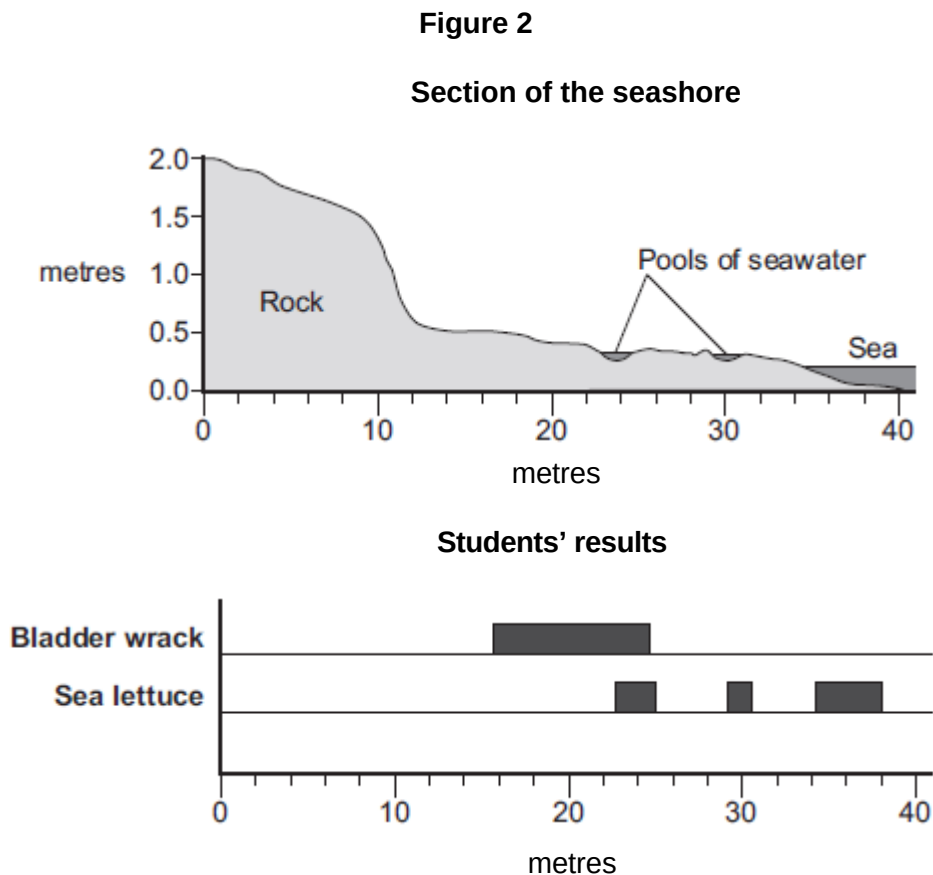


(a) The students:

- 1 placed a 50-metre tape measure on the rocks at right angles to the sea
- 2 placed a quadrat next to the tape measure
- 3 recorded whether each species was present or not.

The students repeated steps 2 and 3 every metre down the shore.

Figure 2 shows a section of the seashore and the students' results.



- (i) The students placed the quadrat at regular intervals along a transect line rather than placing the quadrat at random positions anywhere on the rocky shore.

Explain why.

(2)

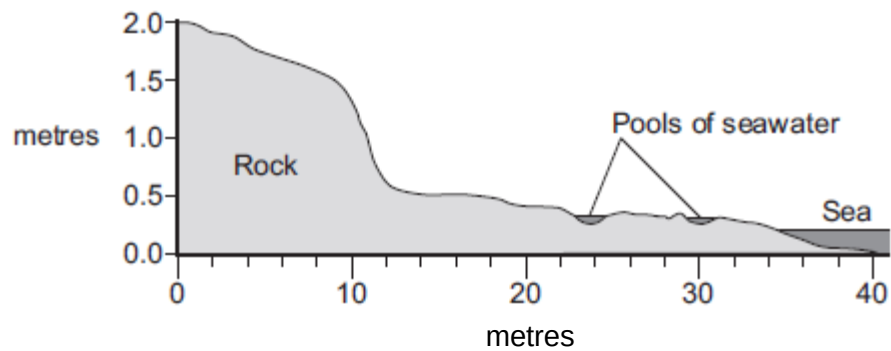
- (ii) How could the students have improved their investigation to ensure that they produced valid data?

(2)

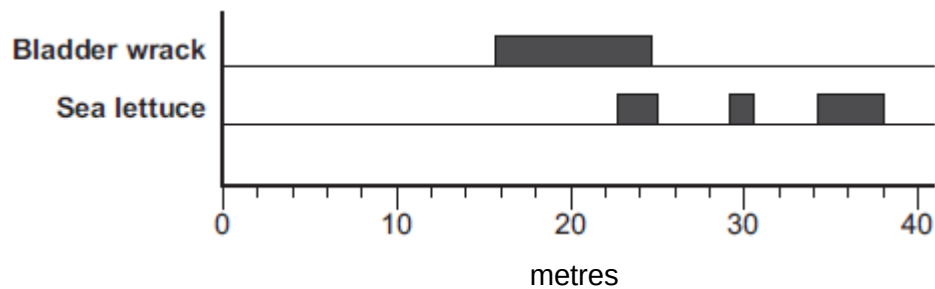
- (iii) **Figure 2** is repeated here to help you answer this question.

Figure 2

Section of the seashore



Students' results



The students concluded that bladder wrack is better adapted than sea lettuce to survive in dry conditions.

What is the evidence for this conclusion?

Use information from **Figure 2**.

(2)

- (b) The bladder wrack has many air bladders.
The air bladders help the bladder wrack to float upwards when the sea covers it.

Suggest how this helps the bladder wrack to survive.

(2)

(Total 8 marks)

3.

Cows are reared for meat production.

The cows can be reared indoors in heated barns, or outdoors in grassy fields.

The table shows energy inputs and energy outputs for both methods of rearing cows.

	kJ / m ² / year		
	Energy input		Energy output
	Food	Fossil fuels	Meat production
Indoors	10 000	6 000	40
Outdoors	5 950	50	X

- (a) The percentage efficiency for rearing cows **outdoors** is 0.03%

Calculate the energy output value **X**.

Use the equation:

$$\text{percentage efficiency} = \frac{\text{energy output}}{\text{total energy input}} \times 100$$

Energy output value **X** = _____ kJ / m² / year

(3)

- (b) The percentage efficiency for rearing cows **outdoors** is 0.03%

Calculate how many times more efficient it is to rear cows indoors than to rear cows outdoors.

Use the equation from (a).

Answer = _____ times

(3)

- (c) A large amount of energy is wasted in both methods of rearing cows.

Give **two** ways in which the energy is wasted.

1. _____

2. _____

(2)

- (d) Suggest **two** reasons why it is more efficient to rear cows indoors than to rear cows outdoors.

1. _____

2. _____

(2)

(Total 10 marks)

4.

A new dog food has been developed that does **not** contain meat from cows, sheep or chickens.

The new dog food contains insects.

The insects in the dog food factory are fed on waste vegetables.

- (a) Sketch the pyramid of biomass for the food chain that produces food for dogs from insects.

Label the pyramid.

(2)

- (b) Describe **two** reasons why the biomass of the insects eaten by dogs does **not** all become biomass of the dogs.

1 _____

2 _____

(2)

- (c) Explain how making dog food from insects could improve **human** food security in the future.

(4)

(Total 8 marks)

5.

Human activities affect the environment.

- (a) Deforestation results in an increase in carbon dioxide levels in the atmosphere.

Give **two** reasons why.

(2)

- (b) *In this question you will be assessed on using good English, organising information clearly and using specialist terms where appropriate.*

A dairy farmer washes out his cow shed each day. The waste water contains urine and faeces. The waste water overflows into a stream by mistake.

The waste water will have an effect on the plants and invertebrates living in the stream.

Explain why.

[illegible]

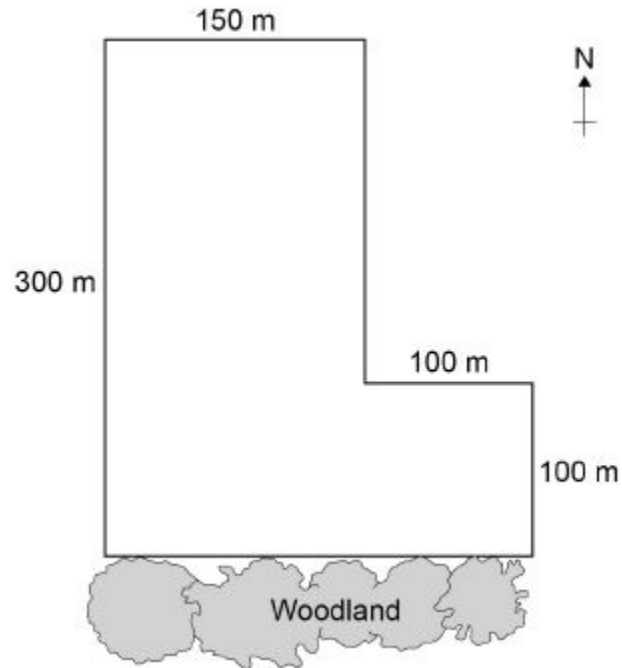
(6)

(Total 8 marks)

6.

Some students investigated the size of a population of dandelion plants in a field.

The diagram below shows the field.



The students:

- placed a 1 m × 1 m square quadrat at 10 random positions in the field
- counted the number of dandelion plants in each quadrat.

The table below shows the students' results.

Quadrat number	Number of dandelion plants
1	6
2	9
3	5
4	8
5	0
6	10
7	2
8	1
9	8
10	11

- (a) Why did the students place the quadrats at random positions?

(1)

- (b) Estimate the total number of dandelion plants in the field.

Calculate your answer using information from the diagram and the table above.

Give your answer in standard form.

Total number of dandelion plants = _____

(5)

(d) Light is an environmental factor that affects the growth of dandelion plants.

Give **two** other environmental factors that affect the growth of dandelion plants.

1. _____

2. _____

(2)

(Total 14 marks)

Mark schemes

1.

(a) primary consumer

1

(b) correct shape: 4 tiers with largest at bottom and smallest at top

1

correctly labelled:

dragonfly / nymph

+ hydra

+ daphnia

+ algae

in this order

or allow:

3rd-order or tertiary consumer or apex / top predator or (trophic level) 4

2nd-order or secondary consumer or (trophic level) 3

*1st-order or primary consumer or herbivore or (trophic level) 2 producer
or (trophic level) 1*

allow for 2 marks inverted pyramid if correctly labelled

1

(c) any **one** from:

(Daphnia biomass smaller because)

- non-digestible parts (of algae) or lost in faeces

ignore waste

- not all absorbed
- lost in urine / urea
- used in respiration **or** lost as carbon dioxide / CO₂
allow excretion
allow (to supply energy) for movement / warmth
allow used to supply energy
- algae not all eaten **or** eaten by other organisms
- some algae decompose

1

(d)

an answer of 14 000 scores 2 marks

14

1

14 000

allow evidence of an incorrectly calculated mean $\times 1000$

allow 1.4×10^4

1

(e)

an answer of 2.625×10^4 **or** 2.63×10^4 **or** 2.6×10^4
scores **4** marks

an answer of 26250 scores **3** marks

allow ecf from part **(d)**

(volume of pond =) 1.875 **or** $2.5 \times 1.5 \times 0.5$

an incorrect answer for one step does **not** prevent
allocation of marks for subsequent steps

1

$14\,000 \times 1.875$

allow ecf from part **(d)**

1

26250

1

2.625×10^4

allow 2.63×10^4 **or** 2.6×10^4

1

(f) increased (growth / reproduction of) algae

1

(more algae so) more food for Daphnia

allow fertiliser toxic to Hydra (1) (so) fewer Daphnia
eaten (1)

1

(g) (Hydra have) less food

1

because (graph shows) fewer Daphnia (with more fertiliser)

allow other valid suggestions, eg
fertiliser toxic to Hydra (1)

or

fertiliser causes growth of algae (on surface) which
block light and so die and decay

or

eutrophication (1)

(decay / eutrophication) uses up oxygen (so lack of
oxygen for Hydra) (1)

1

[14]

2.

(a) (i) to get data re position of seaweed / of organism

1

in relation to distance from sea / distance down shore / how long each seaweed
was exposed

1

- (ii) repeat several times
minimum = 2 repeats

1

elsewhere along the shore

1

- (iii) bladder wrack is further up the shore (than the sea lettuce) / exposed for longer
ignore found in dry areas / on bare rock

1

sea lettuce (only) in rock pools / in the sea / (only) in water

1

- (b) gets more light / closer to light
allow better access to CO₂

1

- (so) more photosynthesis
allow 1 mark for light for photosynthesis
allow 1 mark for CO₂ for photosynthesis
ignore reference to oxygen for respiration
'more' only needed once for 2 marks

1

[8]

3.

- (a) $0.03 = \frac{\text{output}}{5950 + 50} \times 10$
an answer of 1.8 scores 3 marks

1

$$\text{output} = \frac{0.03 \times (590 + 50)}{100}$$

1

1.8

1

- (b) indoor % efficiency = $\frac{40}{10000 + 6000} \times 100$

1

or

$$\frac{40}{16000} \times 100$$

0.25(%)

an answer of 8.33 scores 3 marks
allow 8 / 8.3 / 8.333...

1

$$\left(\frac{0.25}{0.03} = \right) 8.33 \text{ (times)}$$

1

(c) any **two** from:

- in faeces / egestion
- or
- not all food is absorbed
- not all food is ingested
- in urine / excretion
- in respiration
- keeping warm
- movement

*do **not** accept 'for respiration'*

allow as 'heat'

2

(d) warmer indoors so less energy wasted in keeping warm

allow less energy lost as 'heat'

1

less movement indoors so less energy wasted

*if no other mark awarded, allow it is warmer and there is less movement indoors for **1** mark*

1

[10]

4.

(a) triangular pyramid with 3 levels

1

correct labels: (waste) vegetables / plants; insect(s); dog(s)

*do **not** accept additional incorrect labels*

1

(b) any **two** from:

- carbon dioxide from respiration (from dog)
allow carbon dioxide breathed out (by dog)
- urea from excretion (from dog)
allow urea in urine (from dog)
- not all parts (of insects) are absorbed / digested (by dog)
allow faeces from egestion (from dog)
ignore references to loss of energy
*if no other mark awarded allow **two** factors without descriptions for **1** mark*

2

(c) less land required

1

(so) more space for crops (for humans)

allow more meat (from cows etc) for humans

1

less methane (from animals) therefore less global warming

allow less methane from rotting vegetables in landfill

1

(therefore) less harmful effects of global warming on (human) food production

allow example such as less flooding of farmland

allow may lead to the development of more foods for humans made from insects

1

[8]

5.

(a) any **two** from:

- fewer trees to take in carbon dioxide for photosynthesis
 - decomposers / microorganisms respire (as they decay debris) releasing carbon dioxide
 - burning of wood releases carbon dioxide
- allow carbon dioxide released by burning fossil fuels in vehicles / factories*

2

- (b) Marks awarded for this answer will be determined by the Quality of Communication (QC) as well as the standard of the scientific response. Examiners should also refer to the information on page 5, and apply a 'best – fit' approach to the marking.

0 marks

No relevant content.

Level 1 (1 – 2 marks)

There is a brief description of some steps in the process but the order is not clear with little biological vocabulary used.

Level 2 (3 – 4 marks)

There is a reasonably clear description of the process involving many of the steps and using some biological vocabulary.

Level 3 (5 – 6 marks)

There is a clear, logical and detailed scientific description of the process using appropriate biological vocabulary.

examples of biology points made in the response:

- this contains mineral ions (and organic matter)
- this increases growth of algae / water plants
- the plants / algae (underneath) die
- due to lack of light / photosynthesis / space
- decomposers / microorganisms feed on decaying matter **or** multiply rapidly
- the respiration of decomposers uses up all the oxygen
- so invertebrates die due to lack of oxygen
- this is called eutrophication

6

[8]

6.

- (a) there is an uneven distribution of dandelions

or

(more) representative / valid

or

avoid bias

or

more accurate / precise mean

ignore accurate / precise unqualified

ignore repeatability / reproducibility / reliability / fair test

1

(b) (correct mean per m² => 6 or 6.0

1

(correct field area => 55 000 (m²)

1

mean × area – e.g. 6(.0) × 55 000

allow incorrect calculated values for mean and / or field area

1

330 000

allow correct calculation from previous calculation

1

3.3×10^5

allow calculated value in standard form

1

an answer of 3.3×10^5 scores 5 marks

an answer of 330 000 scores 4 marks

(c) **Level 3:** The method would lead to the production of a valid outcome. All key steps are identified and logically sequenced.

5–6

Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the method is not fully logically sequenced.

3–4

Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.

1–2

No relevant content

0

Indicative content

- placing of quadrat
- large number of quadrats used
- how randomness achieved – e.g. table of random numbers **or** random number button on calculator **or** along transect
- quadrats placed at coordinates **or** regular intervals along transect
- in each of two areas of different light intensities **or** transect running through areas of different light intensity
- for each quadrat count number of dandelions
- for each quadrat measure light intensity
- compare data from different light intensity

to access **level 3** the key ideas of using a large number of quadrats randomly, or along a transect, and counting the number of dandelions in areas of differing light intensity need to be given to produce a valid outcome

- (d) any **two** from:
- temperature
allow heat
 - water
allow moisture / rain
 - (soil) pH
allow acidity
 - minerals / ions
*allow e.g. magnesium ions **or** nitrate*
allow salts / nutrients
 - winds
 - herbivores
allow trampling
ignore carbon dioxide
ignore space
ignore competition unqualified
*do **not** accept oxygen*

2

[14]

Examiner reports

1.

- (a) Three-quarters of students correctly selected 'primary consumer' as the term to describe the Daphnia. 'Secondary consumer' was the next most common option.
- (b) More than half of the students were able to draw and correctly label a pyramid of biomass for the food chain. Some drew the pyramid upside down, which was acceptable if the levels were appropriately labelled. Others inverted the labels and so did not gain the mark.

Many added multiple labels which was acceptable provided every form of labelling was completely correct. For example, the Daphnia could be called by its name or called a 'primary consumer', '1st-order consumer', 'herbivore' or 'trophic level 2'; some students did not gain the mark due to a labelling error.

- (c) Vague answers given by some students - 'because biomass is lost between trophic levels' or 'because there were fewer Daphnia than algae' were not sufficient to gain credit. The mechanism of loss of biomass was required e.g. non-digestible material/used in respiration/the algae being eaten by other organisms. About 7% of students gave a creditworthy answer.
- (d) In this question, students had to calculate the mean of five counts of Daphnia, each from a 1 dm^3 sample of pond water and, given that $1 \text{ m}^3 = 1000 \text{ dm}^3$, scale the value up to a mean number in 1 m^3 of pond water. The actual mean of the five samples was 14 and so an answer of 14 000 in 1 m^3 should have been calculated.

Errors were very prevalent: some divided by 1000 and so obtained an impossible value of 0.014 Daphnia per m^3 . Others omitted the value of zero given in the table and so calculated a value of 17 500 per m^3 . There were also various arithmetic errors among those who first multiplied each of the five values in the table by 1000 before finding the mean. More than a fifth of students gave the correct final answer.

- (e) Given the dimensions of the pond, students had to use the value obtained in part (d) to calculate the estimated number of Daphnia in the whole pond, with their final answer in standard form. Since the dimensions of the pond were given in metres, there was no need for any conversion of units – but this did not prevent many students trying to do so. The actual calculation was: answer from part (d) $\times$ volume of pond.

However, some miscalculated the volume of the pond, dividing rather than multiplying and many did not present their final answer in standard form. There was a relatively even split of students across the mark range, with about 15% gaining full credit. This question did differentiate between students very well.

- (f) The data in Figure 2 showed that a large number of Daphnia were found at the given concentration of fertiliser in the pond water. To account for this, students had to understand that the fertiliser would have been used by the algae and therefore the number of algae would have increased, thus providing more food for the Daphnia. Many students thought that the Daphnia fed directly on fertiliser or that the fertiliser provided 'optimum conditions' for the Daphnia. Around 15% of students scored any marks for this question.

- (g) The situation here paralleled that in the previous question except that a decrease in the population of the Hydra had to be explained. Since the specified concentration of fertiliser had resulted in a low number of Daphnia, students were expected to deduce that the Hydra would not have enough to feed on and hence its numbers would decline.

Some students thought incorrectly in terms of absolutes: 'The hydra would have no daphnia to eat.' An alternative explanation in terms of the high concentration of fertiliser possibly being toxic to the Hydra was also accepted. A significant number of students appeared to believe that Hydra is a plant and that lack of sunlight (caused by a layer of fertiliser or algae) would 'prevent it from photosynthesising'. Students were generally more successful in this question than in part (f).

2.

- (a) (i) The techniques of random sampling with quadrats and sampling along a transect are both included in section B2.4.1(b) of the Specification. Students were required to explain why, when investigating 'whether two different species of seaweed could live only at certain positions on a rocky shore', the transect method was preferable. Many students decided that this was the time to make use of the terminology they associated with investigations and so wrote about one or more of 'fair / valid / reliable / reproducible / precise / accurate', all of which were inappropriate. Some even thought that the random quadrat method might be 'biased'. Thus more than half of the students scored zero marks. The results given in the diagram also showed that the positions at which data were collected related to the distance down the shore towards the sea.
- (ii) Many students gave the simple response of 'repetition' as an improvement to ensure more valid data. Examiners expected that such repetition should be done several times (at least twice more) and at different positions along the shore in order to ensure that the data gathered were more representative than just a one-off set of measurements.
- (iii) Two-thirds of students noticed that the sea lettuce was found further down the shore than the bladder wrack, or that it was always associated with standing water – either in pools or in the sea. However, only a small minority were able to make both of these points. Many also failed to make their answers comparative, for example stating that the bladder wrack was found between 15 and 25 metres on the transect line but not pointing out that this was further up the shore than most of the sea lettuce.
- (b) The stem to the whole of question 5 had stated that 'Seaweeds are plant-like organisms that make their food by photosynthesis.' Less than half of the students were able to relate the upward floating of the bladder wrack, assisted by its bladders full of air, to an improved ability to photosynthesise as it was more exposed to light. Some ignored the fact that the seaweed was fixed to the rock (again, given in a diagram near the beginning of the question) and envisaged the seaweed free-floating on the surface of the water where it could 'breathe' the air or drift off to find new shores to colonise.

4.

This question presented students with a novel situation: the development of a new type of food for dogs from insects. This was set in the context of a food chain, the transfer of biomass and, ultimately, in the provision of food security for humans. Part (a) was a relatively straightforward question for students. Problems mainly arose in the labelling with the use of terms such as 'secondary' and 'tertiary' consumers (the latter being entirely inappropriate), or by adding multiple labels that included the names of the organisms, the types of producer and consumer and the numbered trophic levels, but with an error in one of these, and thus this mark could not be awarded. In (b) reasons given by students for the incomplete transfer of biomass from the insects to the dogs were largely incorrect, with 1% of students being able to describe two. Some students made inappropriate references to some of these processes occurring in the insects rather than in the dogs.

Part (c) related the scenario of insect-based dog food to human food security. Students found this difficult and, although nearly three-quarters scored one mark (usually for stating that more meat would be available for human consumption if dogs were fed on insects), 7% gained further credit. Few realised that if less meat were consumed by dogs, then less land would be required for animal grazing and hence it could be utilised for growing crops. Hardly any suggested that a reduction in farm animals would perhaps reduce methane output and the negative effects of methane on the climate and hence on the growth of crops.

5.

(a) Students were asked to state why deforestation increases carbon dioxide in the atmosphere. Most students scored at least 1 mark on this question, although marks were lost for a lack of detail in responses especially in the first marking point. Students needed to indicate that *fewer* trees were available to photosynthesise and take in carbon dioxide.

Few students used the respiration of decomposers as an example. But many gained a mark from a reference to combustion. The alternative point required students to refer to burning of fossil fuel in the use of vehicles / factories to be awarded the mark.

Another common error was the confusion of photosynthesis and respiration, and some students wrote 'carbon' where they meant carbon dioxide missing out on the mark point. Several mentioned removal of carbon sinks but did not link this into carbon dioxide release or intake.

(b) There were some excellent level 3 responses in this item and many students scored highly on this question. However, there was also a surprising number, almost 25%, who demonstrated no understanding of eutrophication at all. Many could make most of the points from the mark scheme and link them well, but there were a few common errors. The dark / turbid colour of the organic matter was sometimes claimed to be responsible for the lack of light but students could still pick up the remaining points on the mark scheme to gain credit. Some described bacteria in sewage lowering the oxygen level immediately and so many organisms could not survive. This was not creditworthy.

Where major errors were made they were mostly in the understanding of the role of respiration in relation to the plants and decomposers. Frequently the roles of the two organisms were confused. A few students wrote about the problem of disease in the water which was incorrect.

6.

(a) 57% of students understood that the purpose of placing quadrats at random was to:

- allow for the uneven distribution of the dandelion plants
- get a more representative or valid result
- avoid bias
- obtain a more accurate or precise mean value from the 10 samples.

Many others stated that it was to make the results accurate, precise, reproducible, repeatable or fair and such statements were ignored by examiners.

(b) Calculation of the dandelion population in the L-shaped field was a 5-step process, with each step worth a mark:

1. calculation of the mean number of dandelions in 1 m^2 from the 10 results in the table ($6 / \text{m}^2$)
2. calculation of the area of the field ($55\,000 \text{ m}^2$)
3. multiplying the mean by the area ($6 \times 55\,000$)
4. getting the correct answer from step 3 ($330\,000$)
5. converting the answer from step 4 to a number in standard form (3.3×10^5).

Errors made at each step were allowed for in subsequent steps to avoid excessive loss of marks when students clearly understood the method they had to employ. Many students set out their working in a somewhat disorganised way which was difficult for examiners to follow.

44% of students achieved full marks, with 29% gaining four out of the five marks, primarily because they forgot to convert their final answer to a number in standard form.

(c) Many students produced excellent answers to this question. But a large proportion of the remainder did not achieve any marks at all as they chose an inappropriate investigation: sowing dandelion seeds in pots and seeing how well they grew with lights of different intensities shining on them.

Higher-attaining students described how they would run a line transect from the shady woodland across the field, placing the quadrat at regular intervals along this and counting the dandelion plants in each quadrat. They anticipated that the density of dandelions would increase further away from the woodland.

The highest-attaining students stated that a light reading should be taken with a light meter within each quadrat, that the transect should be repeated at several other similar locations, means being calculated for set distances from the woodland, and that a graph should be plotted of the number of dandelions against light intensity, hoping to find a positive correlation. A suitable alternative to this method was to place a large number of random quadrats in a shady area and in a brightly-lit area and determining the mean number of dandelions for each area which could then be compared.

Answers were marked using a level of response mark scheme with:

- level 3 (5–6 marks) being awarded if the method described would give a valid outcome and the key steps were logically sequenced
- level 2 (3–4 marks) for a sensible method which would not necessarily produce a valid outcome as one or two key steps had been omitted.
- level 1 (1–2 marks) giving an answer that would definitely not give a valid outcome, although some relevant material had been included.

Overall, 33% of students achieved level 3.

- (d) Suitable suggestions of two other environmental factors that might have determined how well the dandelions grew included water, mineral ions and the pH of the soil, as well as biotic factors such as herbivores or trampling. 97% of students were able to give at least one factor with 44% managing to provide two.

Factors considered inappropriate included the concentration of carbon dioxide or oxygen in the air, as they were unlikely to be limiting in an open field, but these were often stated by students.